

ALY 6010 Prob Theory and Intro Stat – Report

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Module 5 R Practice

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Introduction:

In the rapidly evolving real estate landscape, grasping the determinants of house prices holds significant importance for investors, developers, and homeowners. Given the growing complexities of property appraisal, harnessing data analytics has become indispensable for comprehending the nuances of housing markets. In this regard, our analysis delves into the assessment of real estate assets, with the objective of revealing the fundamental forces shaping house prices.

Problem Statement:

The analysis aimed to explore the factors influencing house prices and understand the relationships between various variables in the real estate market. Specifically, the study sought to investigate the impact of house age on house price per unit area and identify other relevant factors contributing to property valuation. By examining correlations and building a linear regression model, the goal was to gain insights into the dynamics of the real estate market and provide valuable information for stakeholders such as homebuyers, sellers, and real estate investors.

Data Used:

Reading data : Importing the file on dataset in R and reading dataset

```
Real_estate <- read_excel("Real estate valuation data set.xlsx")
head(Real_estate)

## # A tibble: 6 × 8
##       No `X1 transaction date` `X2 house age` X3 distance to the nearest MR
T st...1
##       <dbl>                <dbl>          <dbl>
<dbl>
## 1         1              2013.             32
84.9
## 2         2              2013.             19.5
307.
## 3         3              2014.             13.3
562.
## 4         4              2014.             13.3
562.
## 5         5              2013.              5
391.
## 6         6              2013.             7.1
2175.
## #  abbreviated name: 1`X3 distance to the nearest MRT station`
## #  4 more variables: `X4 number of convenience stores` <dbl>,
## #       `X5 latitude` <dbl>, `X6 longitude` <dbl>,
## #       `Y house price of unit area` <dbl>
```

Installing libraries

```
library(readxl)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(reshape2)
library(corr)
library(ggplot2)
library(tidyverse)

## — Attaching core tidyverse packages ————— tidyverse 2.
0.0 —
## ✓ forcats   1.0.0      ✓ stringr   1.5.1
## ✓ lubridate 1.9.3      ✓ tibble    3.2.1
## ✓ purrr     1.0.2      ✓ tidyr     1.3.0
## ✓ readr     2.1.5

## — Conflicts ————— tidyverse_conflict
s() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force al
l conflicts to become errors

library(broom)
```

Explore the structure of the data

Structuring the dataset to examine its characteristics, including the variables it contains, their types, the number of observations, and any missing values or outliers.

```
str(Real_estate)

## tibble [414 × 8] (S3: tbl_df/tbl/data.frame)
##   $ No                                     : num [1:414] 1 2 3 4 5 6 7 8 9 10 .
##   ..
##   $ Transaction date                     : num [1:414] 2013 2013 2014 2014 20
13 ...
##   $ House age                           : num [1:414] 32 19.5 13.3 13.3 5 7.
1 34.5 20.3 31.7 17.9 ...
##   $ Distance to the nearest MRT station: num [1:414] 84.9 306.6 562 562 390
```

```
.6 ...
## $ No. of Convenience store      : num [1:414] 10 9 5 5 5 3 7 6 1 3 .
..
## $ Latitude                      : num [1:414] 25 25 25 25 25 ...
## $ Longitude                    : num [1:414] 122 122 122 122 122 ..
.
## $ House price of unit area      : num [1:414] 37.9 42.2 47.3 54.8 43
.1 32.1 40.3 46.7 18.8 22.1 ...
```

Data Types:

No: Numeric (continuous)

Transaction date: Numeric (continuous)

House age: Numeric (continuous)

Distance to the nearest MRT station: Numeric (continuous)

No. of Convenience store: Numeric (continuous)

Latitude: Numeric (continuous)

Longitude: Numeric (continuous)

House price of unit area: Numeric (continuous)

Observations:

There are 414 observations in total.

Data Cleaning and Transformation:

Data cleaning and transformation are crucial steps in the data analysis process, aimed at ensuring that the dataset is accurate, complete, and formatted appropriately for analysis.

Renaming column:

The dataset's columns have been renamed to enhance clarity and understanding. This involved assigning new names to the various variables in the data. By doing so, it simplifies the process for users to comprehend the meaning of each part of the dataset. These alterations improve clarity and ensure that individuals can interpret and analyze the information without confusion.

```
Real_estate <- Real_estate %>%
  rename(`Transaction date` = `X1 transaction date`,
         `House age` = `X2 house age`,
         `Distance to the nearest MRT station` = `X3 distance to the nearest
MRT station`,
         `No. of Convenience store` = `X4 number of convenience stores`,
         `Latitude` = `X5 latitude`,
```

```

      `Longitude` = `X6 longitude`,
      `House price of unit area` = `Y house price of unit area`)
head(Real_estate)

## # A tibble: 6 × 8
##       No `Transaction date` `House age` `Distance to the nearest MRT station`
##   <dbl>         <dbl>         <dbl>         <dbl>
## 1     1           2013.           32             84
## 2     2           2013.          19.5           307
## 3     3           2014.          13.3           562
## 4     4           2014.          13.3           562
## 5     5           2013.           5             391
## 6     6           2013.           7.1           2175
## # 4 more variables: `No. of Convenience store` <dbl>, Latitude <dbl>,
## # Longitude <dbl>, `House price of unit area` <dbl>

```

Checking for missing value

Checking for missing values in a dataset is essential to facilitate data preprocessing and ensure data quality.

```

summary(is.na(Real_estate))

##       No       Transaction date House age
## Mode :logical Mode :logical   Mode :logical
## FALSE:414    FALSE:414        FALSE:414
## Distance to the nearest MRT station No. of Convenience store Latitude
## Mode :logical                               Mode :logical       Mode :logical
## FALSE:414                               FALSE:414           FALSE:414
## Longitude       House price of unit area
## Mode :logical   Mode :logical
## FALSE:414       FALSE:414

```

Identifying class of Dataset

```

Real_estate.class <- sapply(Real_estate, class)
Real_estate.class

##              No              Transaction date
##      "numeric"      "numeric"
## House age Distance to the nearest MRT station
##      "numeric"      "numeric"

```

```
##           No. of Convenience store           Latitude
##           "numeric"           "numeric"
##           Longitude           House price of unit area
##           "numeric"           "numeric"
```

Explanatory Data Analysis (EDA):

Conducting a thorough EDA to understand the distributions, relationships, and trends within the data. Using visualizations such as scatter plots, histograms, and box plots to examine the relationships between the dependent and independent variables.

Training dataset

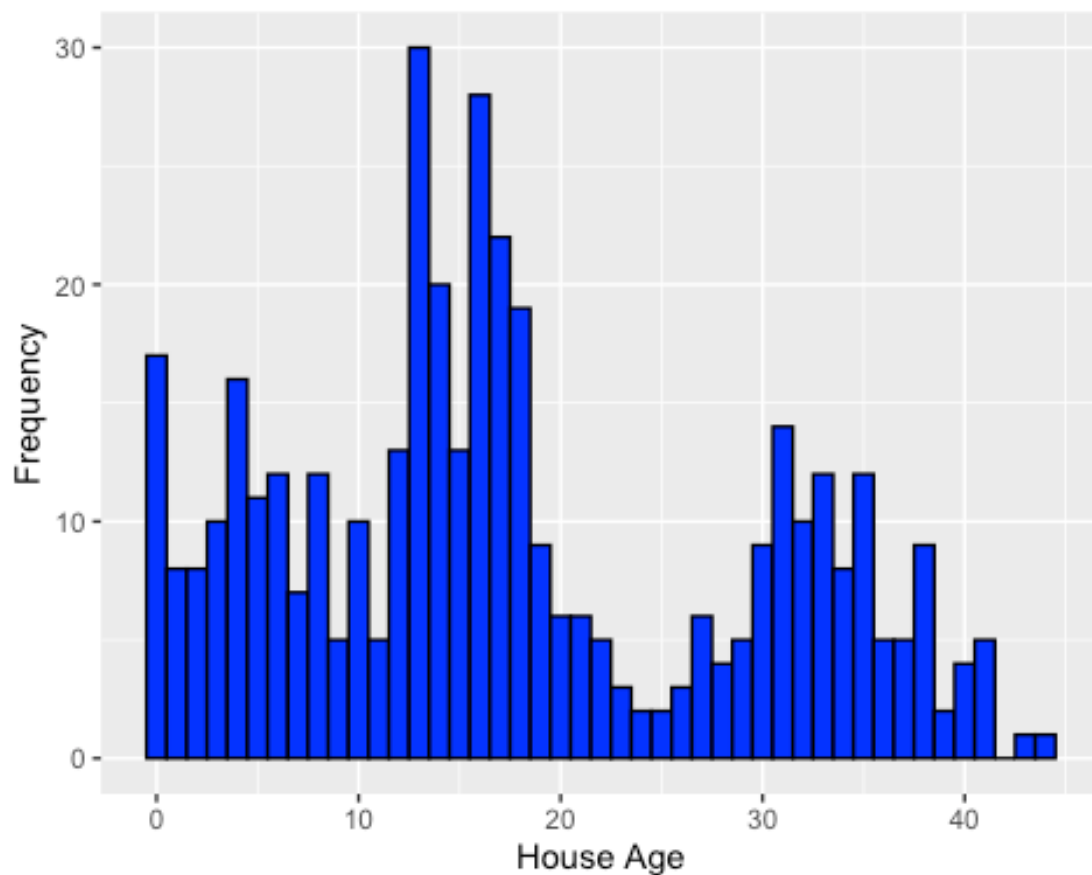
A training dataset is required in machine learning and statistical modeling for several reasons: Such as, Model Development, Model Evaluation and to identify important features.

```
trainingSet = read_excel("Real estate valuation data set.xlsx")
head(trainingSet)

## # A tibble: 6 × 8
##       No `X1 transaction date` `X2 house age` X3 distance to the nearest MR
T st...1
##       <dbl>           <dbl>           <dbl>
<dbl>
## 1         1           2013.           32
84.9
## 2         2           2013.           19.5
307.
## 3         3           2014.           13.3
562.
## 4         4           2014.           13.3
562.
## 5         5           2013.            5
391.
## 6         6           2013.           7.1
2175.
## #  abbreviated name: 1`X3 distance to the nearest MRT station`
## #  4 more variables: `X4 number of convenience stores` <dbl>,
## # `X5 latitude` <dbl>, `X6 longitude` <dbl>,
## # `Y house price of unit area` <dbl>
```

Histogram for House Age

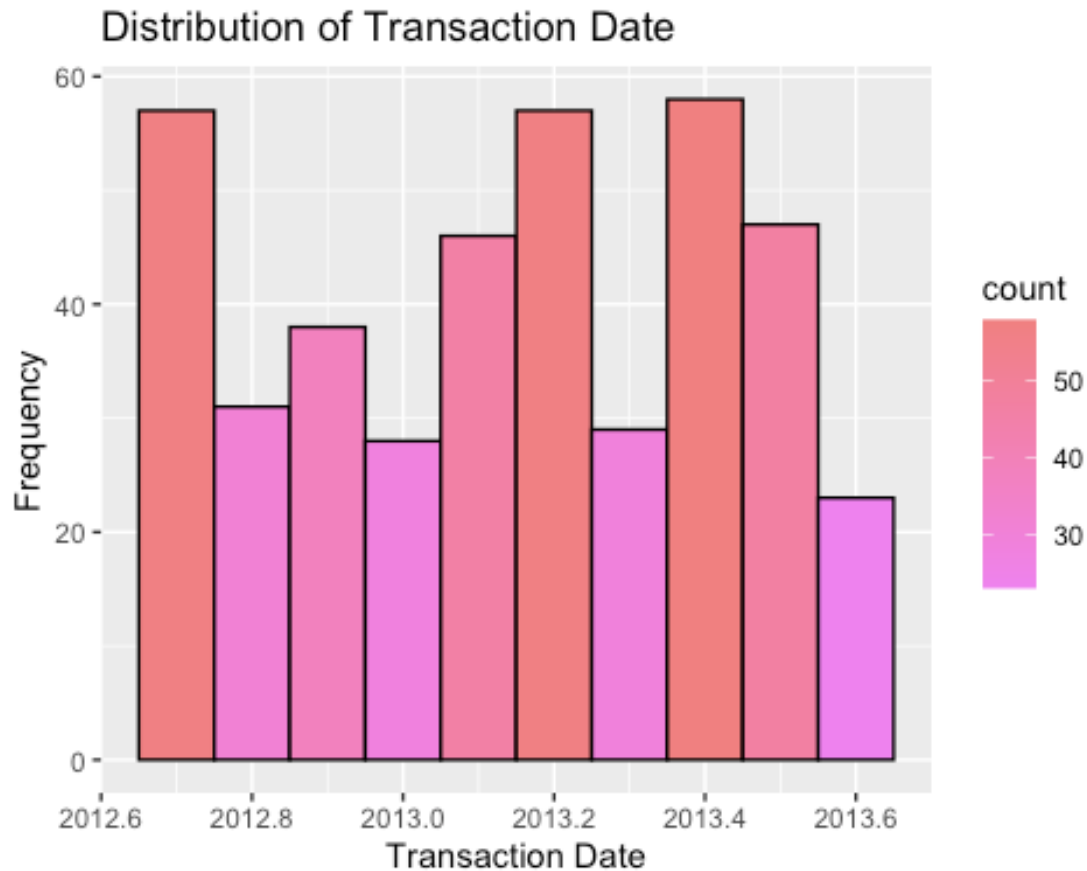
```
ggplot(Real_estate, aes(x = `House age`)) +  
  geom_histogram(binwidth = 1, fill = "blue", color = "black") +  
  labs(x = "House Age", y = "Frequency")
```



Description: A histogram for the "House Age" variable provides a visual representation of the distribution of house ages in the dataset. X-axis (House Age), Y-axis (Frequency) and The shape of the histogram provides insights into the distribution of house ages. Notably, the histogram reveals that the majority of houses fall within the age range of 10 to 15 years, indicating a concentration of properties within this particular age bracket.

Distribution of Transaction Date

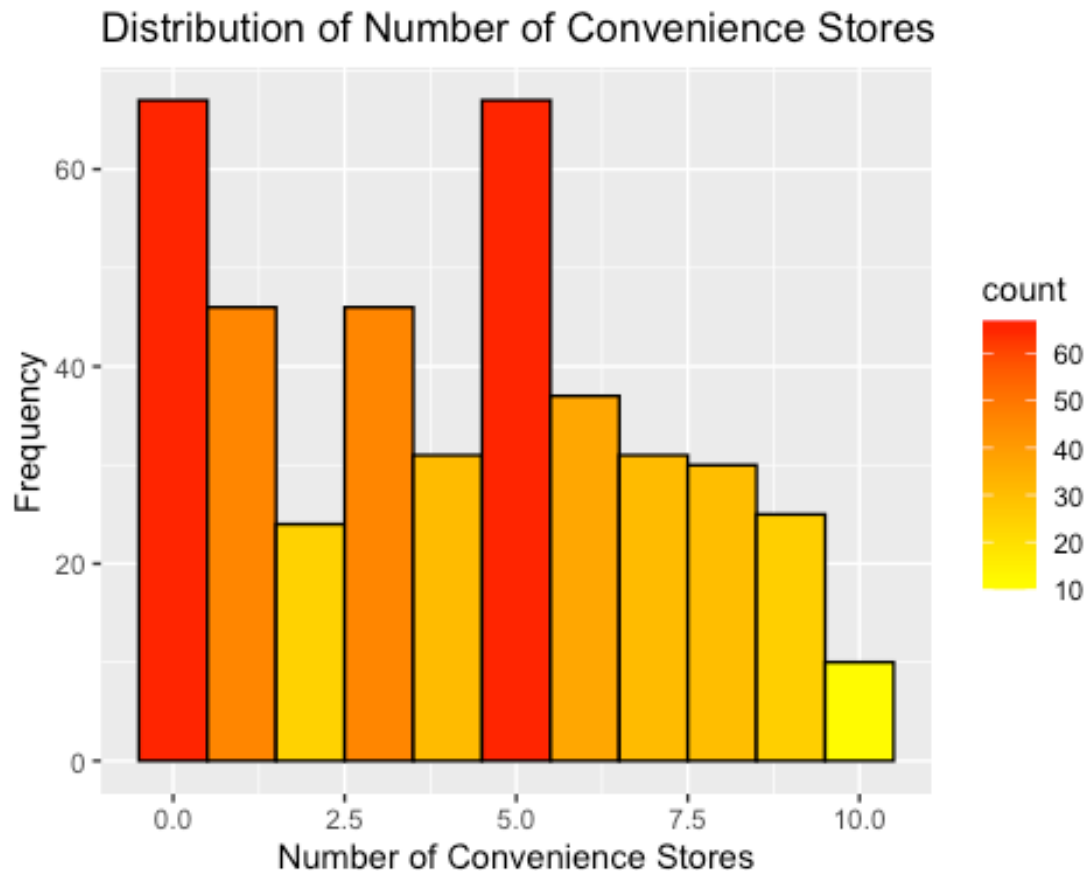
```
ggplot(Real_estate, aes(x = `Transaction date`, fill = after_stat(count))) +  
  geom_histogram(binwidth = 0.1, color = "black") +  
  labs(x = "Transaction Date", y = "Frequency") +  
  ggtitle("Distribution of Transaction Date") +  
  scale_fill_gradient(low = "violet", high = "lightcoral")
```



Description: The distribution of transaction dates provides a visual overview of when real estate transactions occurred within the dataset. In this histogram, the x-axis represents the range of transaction dates, while the y-axis indicates the frequency or count of transactions that took place during each time period.

Distribution of Number of Convenience Stores

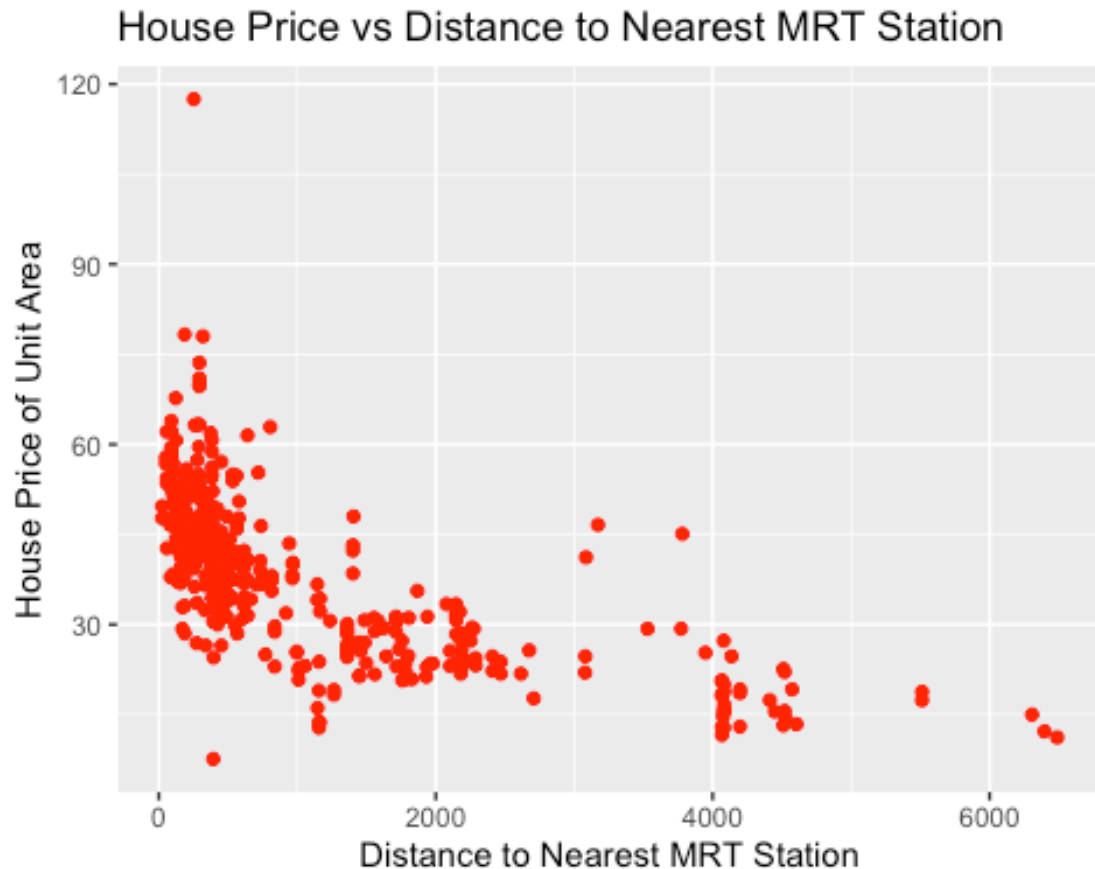
```
ggplot(Real_estate, aes(x = `No. of Convenience store`, fill = after_stat(count))) +  
  geom_histogram(binwidth = 1, color = "black") +  
  labs(x = "Number of Convenience Stores", y = "Frequency") +  
  ggtitle("Distribution of Number of Convenience Stores") +  
  scale_fill_gradient(low = "yellow", high = "red")
```



Description: The histogram illustrates the frequency of properties based on the count of convenience stores in their vicinity. Notably, more than 120 properties are associated with either zero or five convenience stores nearby.

House Price vs Distance to Nearest MRT Station

```
ggplot(Real_estate, aes(x = `Distance to the nearest MRT station`, y = `House price of unit area`)) +  
  geom_point(color = "red") +  
  labs(x = "Distance to Nearest MRT Station", y = "House Price of Unit Area") +  
  ggtitle("House Price vs Distance to Nearest MRT Station")
```



Description: The scatter plot illustrates the correlation between the house price per unit area and the distance to the nearest MRT station, measured in meters. The majority of data points on the plot are concentrated within a distance of 2000 meters from the nearest MRT station. This concentration suggests that most properties in the dataset are located relatively close to MRT stations.

Correlation Coefficient

The correlation coefficient of approximately -0.210567 suggests a weak negative linear relationship between "House age" and "House price of unit area." This means that as the age of the house increases, the price per unit area tends to decrease slightly, and vice versa, although the relationship is not very strong.

```
correlation_coefficient <- cor(Real_estate$`House age`, Real_estate$`House price of unit area`)  
correlation_coefficient  
## [1] -0.210567
```

Question-1

Correlation Data

Correlation data shows the statistical relationship between 5 variables in a dataset. It quantifies the degree to which changes in one variable are associated with changes in another variable. Correlation measures the strength and direction of the linear relationship between variables, typically ranging from -1 to 1.

```
correlation_data <- Real_estate %>%
  select(`House age`,
         `Distance to the nearest MRT station`,
         `No. of Convenience store`,
         Latitude,
         Longitude,
         `House price of unit area`)
correlation_data
```

A tibble: 414 × 6

	<code>`House age`</code>	<code>Distance to the nearest MRT station`</code> <sup>1</sup>	<code>No. of Convenience store`</code> ²	<code>Latitude</code>	<code>Longitude</code>
##	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
##	1	32	84.9	10	25.0
##	2	19.5	307.	9	25.0
##	3	13.3	562.	5	25.0
##	4	13.3	562.	5	25.0
##	5	5	391.	5	25.0
##	6	7.1	2175.	3	25.0
##	7	34.5	623.	7	25.0
##	8	20.3	288.	6	25.0
##	9	31.7	5512.	1	25.0
##	10	17.9	1783.	3	25.0

 404 more rows

 abbreviated names: ¹``Distance to the nearest MRT station``,

²``No. of Convenience store``

 1 more variable: ``House price of unit area`` <dbl>

Computing the correlation matrix

A correlation table is a useful analytical tool that provides insights into the relationships between variables in a dataset.

```
correlation_matrix <- cor(correlation_data)
correlation_matrix
```

##	House age		
## House age	1.00000000		
## Distance to the nearest MRT station	0.02562205		
## No. of Convenience store	0.04959251		
## Latitude	0.05441990		
## Longitude	-0.04852005		
## House price of unit area	-0.21056705		
##	Distance to the nearest MRT station		
## House age	0.02562205		
## Distance to the nearest MRT station	1.00000000		
## No. of Convenience store	-0.60251914		
## Latitude	-0.59106657		
## Longitude	-0.80631677		
## House price of unit area	-0.67361286		
##	No. of Convenience store	Latitude	
## House age	0.04959251	0.0544199	
## Distance to the nearest MRT station	-0.60251914	-0.5910666	
## No. of Convenience store	1.00000000	0.4441433	
## Latitude	0.44414331	1.00000000	
## Longitude	0.44909901	0.4129239	
## House price of unit area	0.57100491	0.5463067	
##	Longitude	House price of unit area	
## House age	-0.04852005	-0.2105670	
## Distance to the nearest MRT station	-0.80631677	-0.6736129	
## No. of Convenience store	0.44909901	0.5710049	
## Latitude	0.41292394	0.5463067	
## Longitude	1.00000000	0.5232865	
## House price of unit area	0.52328651	1.00000000	

The correlation table displays the correlation coefficients between pairs of variables in the dataset. This correlation table is made between 5 variable of data set which are House age, Distance to the nearest MRT station, No. of Convenience store, Latitude, Longitude, House price of unit area.

Convert correlation matrix to long format

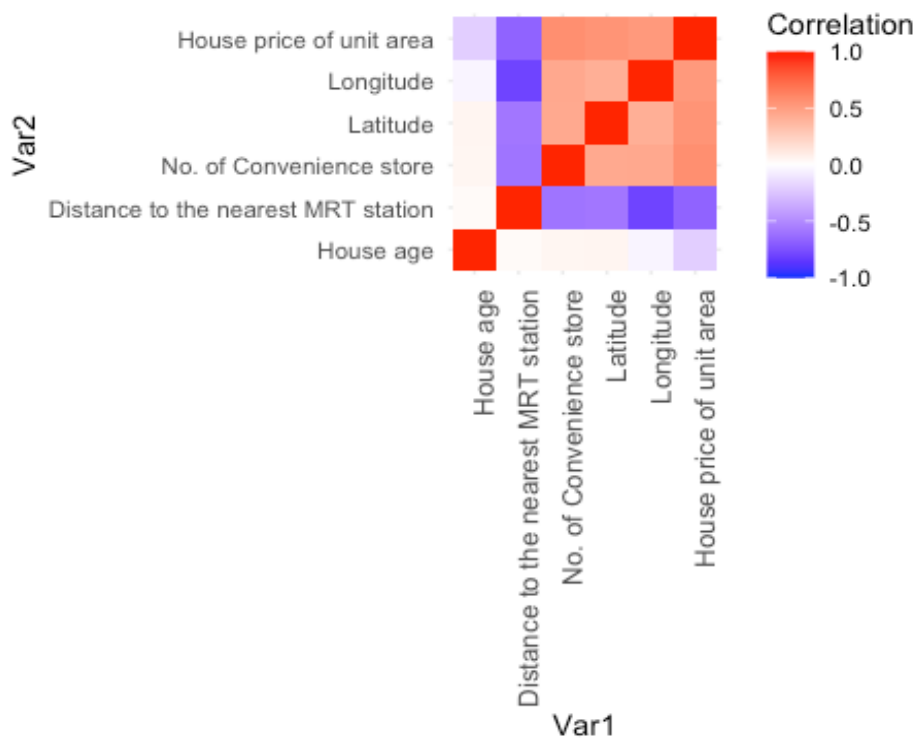
By restructuring the correlation matrix into a long-format dataset, each correlation coefficient is represented as a separate observation, with corresponding variables indicating the pair of variables being correlated and their correlation values.

```
correlation_matrix_long <- melt(correlation_matrix)
head(correlation_matrix_long)
```

##	Var1	Var2	value
## 1	House age	House age	1.00000000
## 2	Distance to the nearest MRT station	House age	0.02562205
## 3	No. of Convenience store	House age	0.04959251
## 4	Latitude	House age	0.05441990
## 5	Longitude	House age	-0.04852005
## 6	House price of unit area	House age	-0.21056705

Plotting heatmap

```
ggplot(correlation_matrix_long, aes(Var1, Var2, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint =
0, limit = c(-1, 1), space = "Lab", name="Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 1, size = 10, hjust =
1)) +
  coord_fixed()
```



Description:

House Age: It has a weak positive correlation with the distance to the nearest MRT station (0.0256), indicating a slight tendency for older houses to be slightly farther from MRT

stations. It also shows weak positive correlations with the number of convenience stores (0.0496), latitude (0.0544), and longitude (-0.0485).

Distance to the Nearest MRT Station: It has a weak positive correlation with house age (0.0256), indicating a slight tendency for houses farther from MRT stations to be older.

It has strong negative correlations with the number of convenience stores (-0.6025), latitude (-0.5911), and longitude (-0.8063), suggesting that houses closer to MRT stations tend to have fewer convenience stores nearby and are located more to the south and west. It also has a strong negative correlation with the house price of unit area (-0.6736), implying that houses closer to MRT stations tend to have higher prices.

Number of Convenience Stores:

It has a weak positive correlation with house age (0.0496), latitude (0.4441), and longitude (0.4491), indicating a slight tendency for areas with more convenience stores to have older houses and to be located more to the north and east. It also has a moderate positive correlation with the house price of unit area (0.5710), suggesting that areas with more convenience stores tend to have higher house prices.

Latitude:

It has weak positive correlations with house age (0.0544), distance to the nearest MRT station (-0.5911), and the number of convenience stores (0.4441), indicating slight tendencies related to the geographic location of the properties. It also has a moderate positive correlation with the house price of unit area (0.5463), suggesting that properties located farther north tend to have higher prices.

Longitude:

It has a weak negative correlation with house age (-0.0485) and weak positive correlations with the number of convenience stores (0.4491) and latitude (0.4129). It has a moderate positive correlation with the house price of unit area (0.5233), indicating that properties located more to the east tend to have higher prices.

House Price of Unit Area:

It has a moderate negative correlation with house age (-0.2106), indicating that older houses tend to have lower prices per unit area. It has strong negative correlations with the distance to the nearest MRT station (-0.6736) and latitude (-0.5463), indicating that properties closer to MRT stations and located farther north tend to have higher prices per unit area. It also has a moderate positive correlation with the number of convenience stores (0.5710), longitude (0.5233), suggesting that areas with more convenience stores and located more to the east tend to have higher prices per unit area. These correlations provide valuable insights into the relationships between the variables in the dataset, which can inform further analysis and decision-making processes.

Question-2

Modeling:

Linear regression model

Performing linear regression model between **House price of unit area** and **House age**.

The predictor variable is "House age."

A predictor variable, also known as an independent variable, is a variable that is used to predict the variation in the outcome variable (also called the dependent variable). In this case, "House age" is the predictor variable because it is used to predict the variation in the "House price of unit area."

```
model <- lm(`House price of unit area` ~ `House age`, data = Real_estate)
model

##
## Call:
## lm(formula = `House price of unit area` ~ `House age`, data = Real_estate)
##
## Coefficients:
## (Intercept)  `House age`
##      42.4347      -0.2515

summary(model)

##
## Call:
## lm(formula = `House price of unit area` ~ `House age`, data = Real_estate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.113 -10.738   1.626   8.199  77.781
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  42.43470     1.21098   35.042 < 2e-16 ***
## `House age`  -0.25149     0.05752   -4.372 1.56e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.32 on 412 degrees of freedom
## Multiple R-squared:  0.04434,    Adjusted R-squared:  0.04202
## F-statistic: 19.11 on 1 and 412 DF,  p-value: 1.56e-05
```

In the context of the model:

Predictor variable ("House age"): This variable represents the age of the house, which is hypothesized to have an effect on the "House price of unit area." The coefficient associated with this variable (-0.2515) indicates the expected change in the outcome variable (house price per unit area) for each one-unit increase in the predictor variable (house age), holding all other variables constant.

Result from the Model

Coefficients:

Intercept: The intercept of 42.4347 represents the estimated average "House price of unit area" when the "House age" is zero. However, this may not be practically meaningful because houses cannot have a negative age.

House age: The coefficient of -0.2515 indicates that for each one-year increase in the age of the house, the predicted "House price of unit area" decreases by approximately 0.2515 units, holding all other variables constant.

Residuals: The residuals represent the differences between the observed and predicted values of the dependent variable ("House price of unit area"). The summary provides statistics about the distribution of these residuals, including minimum, 1st quartile, median, 3rd quartile, and maximum values.

Model Fit:

Residual standard error: The residual standard error of 13.32 indicates the average amount that the observed "House price of unit area" deviates from the predicted values.

Multiple R-squared: The multiple R-squared value of 0.04434 represents the proportion of variance in the dependent variable explained by the independent variable ("House age"). In this model, approximately 4.43% of the variability in "House price of unit area" can be explained by "House age."

Adjusted R-squared: The adjusted R-squared value of 0.04202 adjusts the multiple R-squared value for the number of predictors in the model. It provides a more conservative estimate of the model's goodness of fit.

F-statistic: The F-statistic of 19.11 tests the overall significance of the regression model. The associated p-value (1.56×10^{-5}) indicates that the regression model as a whole is statistically significant.

Interpretation:

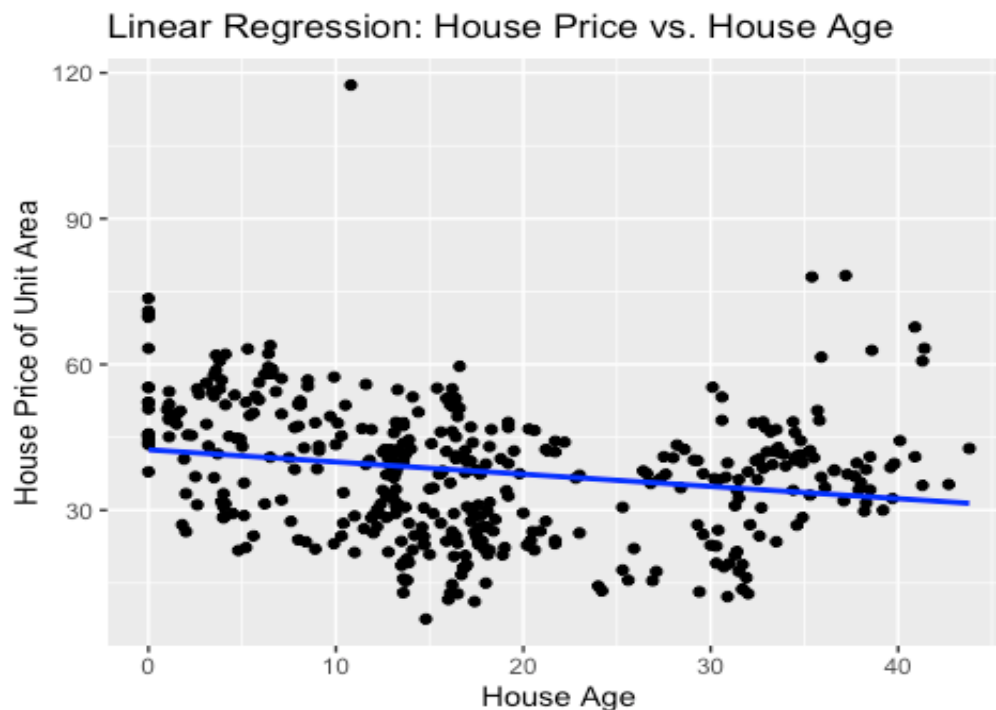
The statistically significant p-value for the coefficient of "House age" (1.56×10^{-5}) suggests that there is a significant relationship between the age of the house and its price per unit area.

However, the low R-squared value indicates that the model explains only a small proportion of the variability in house prices. Other factors not included in the model may also influence house prices.

Overall, while the age of the house has a statistically significant effect on its price per unit area, it is not the only factor determining house prices. Additional variables may need to be considered to build a more comprehensive model.

Plotting the linear regression line

```
ggplot(Real_estate, aes(x = `House age`, y = `House price of unit area`)) +  
  geom_point() + # Scatter plot of the data points  
  geom_smooth(method = "lm", se = FALSE, color = "blue") + # Linear regression  
  on line  
  labs(x = "House Age", y = "House Price of Unit Area") +  
  ggtitle("Linear Regression: House Price vs. House Age")  
## `geom_smooth()` using formula = 'y ~ x'
```



Description: X-axis represents the values of the predictor variable, "House age." Y-axis represents the predicted values of the target variable, "House price of unit area". Each data point from the dataset is represented by a dot on the plot. These dots show the actual values of house prices corresponding to different ages. The line drawn on the plot represents the linear regression model's prediction of house prices based on house age. The slope of the regression line (-0.2515) indicates the rate of change in house price per unit change in house age. The intercept of the regression line (42.4347) represents the predicted house price when the house age is zero.

Conclusion:

The analysis conducted on the real estate valuation dataset provides valuable insights into the factors influencing house prices and the relationships between various variables. Here are the key findings:

The correlation table revealed several interesting relationships between different features. Notably, there was a weak negative correlation between house age and house price per unit area, indicating that older houses tend to have slightly lower prices per unit area. Additionally, variables such as distance to the nearest MRT station, number of convenience stores, latitude, and longitude showed correlations with house price, suggesting spatial and locational influences on property values.

A linear regression model was built to explore the relationship between house age and house price per unit area. The model indicated a statistically significant negative relationship between house age and price per unit area, with older houses generally fetching lower prices. However, the model's low R-squared value suggested that other factors not included in the model may also play a role in determining house prices.